

A Tutorial for Evaluating Cure Model Appropriateness

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ABSTRACT

In survival analysis, traditional models assume all individuals will eventually experience the event of interest. However, advances in therapeutics have led to multiple clinical contexts with potentially curative therapies, and in these contexts, certain individuals may never experience the event. Statisticians have developed cure models as a methodology to address this challenge. Nonetheless, despite significant statistical advances in cure models, we have seen more limited uptake in biomedical applications, and we hypothesize that this is caused by limited guidance in the appropriate application of cure models.

Cure models require specific identifiability conditions for valid parameter estimation, and previous reports have demonstrated significant issues with the inappropriate application of cure models. Existing tutorials for cure models focus on model implementation and either assume or provide only limited guidance on whether cure modeling is appropriate for the given dataset. This tutorial addresses this gap by describing a systematic procedure that integrates clinical judgment, visual inspection of Kaplan-Meier curves, and quantitative evaluation. We provide a worked example using data from a randomized clinical trial in acute myeloid leukemia, and we also summarize findings from a series of other datasets of hematopoietic cell transplantation to suggest broad practical guidance for choosing to apply cure models. By systematically evaluating cure model appropriateness before fitting these models, researchers can achieve more reliable survival analysis and improved clinical decision-making.

KEYWORDS

Cure models, cure fraction, sufficient follow-up time, mixture cure model, survival analysis, oncology

1 | INTRODUCTION

1.1 | Background

Advances in oncology therapies have led to increased long-term survival in certain cancer populations. For example, recent developments in immune checkpoint inhibitors offer potentially improved progression-free survival for patients in early disease stages¹. While multiple myeloma is generally considered incurable, certain subtypes and early-stage cases can achieve long-term remission with chemotherapy, particularly when combined with stem cell transplantation and novel therapies².

Cure models originated in the 1950s⁸ and have since been applied across diverse disciplines from oncology to economics. While traditional models assume that all individuals are susceptible to an event of interest and will eventually experience it, cure models were developed to analyze contexts in which a non-zero fraction of individuals are cured or will never experience the event of interest. When compared to traditional approaches, cure models in appropriate contexts can allow for better prediction of event rates when planning clinical trials³⁴; more efficient test statistics³⁹; and improved estimation in long-term extrapolations⁷.

However, cure models make important assumptions beyond those of traditional survival models. Two key assumptions are that (1) there exists a non-zero proportion of the population that will not experience the event and (2) there exists suitable follow-up in the sample to identify cure model parameters. (The latter condition has received much attention in the literature and is developed with more technical precision in Section 2.1.2.)

Authors have shown that violations to these assumptions can cause significant issues in the results from cure model analysis. Simulation-based studies by Grant et al.³² and Kearns et al.³³ have shown that cure model estimates derived from interim or limited follow-up can be inappropriate compared with estimates obtained at later, more mature follow-up, and that model misspecification further degrades extrapolation performance.

To evaluate the impact of sufficient follow-up time on parameter estimation with cure models in real-world data, Othus et al.¹⁶ analyzed six clinical trials from the SWOG Cancer Research Network each at two different follow-up times. They observed that initial cure model-based estimates of mean survival could meaningfully differ from later estimates, and that the direction of the difference was not predictable. This research demonstrates that estimates from misapplication of cure models can lead to inappropriate results in real-world studies, potentially misleading clinicians and policymakers in evaluating long-term treatment effectiveness.

In recent literature, three tutorials were published on the use of cure modeling, all in the context of health technology assessment (HTA)¹⁸⁻²⁰. Two of them focus on mixture cure modeling and the other one focuses on both mixture and non-mixture cure models. While these tutorials provide comprehensive guidance for applying cure models—including parametric distribution selection by Felizzi et al.¹⁸, excess hazard methods by Sweeting et al.¹⁹, and comparison of mixture versus non-mixture approaches by Latimer and Rutherford²⁰—they do not address a fundamental prerequisite: they do not offer guidance for identifying whether a dataset is likely to be appropriately modeled with cure models (e.g., whether a study’s follow-up is sufficiently long to permit valid cure modeling).

In this work, we aim to address this gap with a tutorial to help users gain a fundamental understanding of the underlying concepts and practical implementation for identifying whether a cure model is appropriate for a given dataset. First, we introduce the data setup and summarize theoretical conditions for considering cure models. Then we discuss methods of visual evidence for cure fraction and sufficient follow-up followed by quantitative methods of assessing cure model appropriateness. Next, we demonstrate a step-by-step process of assessing cure model appropriateness using a worked example. Finally, we describe other practical considerations for applying cure models based on a series of datasets from hematopoietic cell transplantation.

2 | METHODS

2.1 | Mixture Cure Models

For each subject $i = 1, \dots, n$, let T_i denote the (possibly improper) failure time. Subjects are observed under an independent censoring time U_i , and the observed data consist of (Y_i, δ_i) , where $Y_i = \min(T_i, U_i)$ is the observed time and $\delta_i = I(T_i \leq U_i)$ is the event indicator. We assume the failure times T_i are independent and identically distributed (i.i.d.) with cumulative distribution function $F(t) = P(T \leq t)$ and survival function $S(t) = P(T > t)$, $t \geq 0$. In a cure setting, these distribution and survival functions can be improper, with $\lim_{t \rightarrow \infty} S(t) > 0$ and $\lim_{t \rightarrow \infty} F(t) < 1$, reflecting a non-zero probability that the event of interest is never observed.

In this tutorial, we concentrate on mixture cure models^{8,23}, which write the population survival function as a mixture of cured and uncured sub-populations. Let the probability of being in the cured sub-population be $(1 - p)$, and the survival function for the uncured sub-population be the proper function $S_0(t)$. The mixture cure model is then:

$$S(t) = (1 - p) + p \cdot S_0(t), \quad t \geq 0 \quad (4)$$

Note that this formulation models cure status as a latent property of the population: an individual's cure status is not directly observed, and the model is best understood as describing the proportions of long-term and short-term survivors in the population rather than as a tool for classifying individual subjects.

2.1.1 | Practical Considerations when the Outcome of Interest Includes Mortality

In oncology, many time-to-event analyses focus on mortality endpoints, where the literal notion of being “cured” or “immune” to death is unrealistic. Even for event-free survival endpoints, individuals may avoid some components of a composite outcome (e.g., disease relapse) but not death itself. In these settings, cure models are best understood as capturing long-term survivorship rather than absolute immunity: the cure fraction is interpreted as the proportion of the population whose event hazard becomes negligible after a sufficient duration of follow-up. A cure model can therefore still fit the data well when the long-term sub-population is unlikely to experience the event during the observed follow-up. Additionally, the class of relative survival

models⁴⁰ is an alternative class that explicitly accounts for background mortality, if needed to address the residual hazard due to background mortality.

2.1.2 | Identifiability Conditions for Cure Models

Cure models require identifiability conditions beyond those of conventional survival models. “Identifiability” concerns whether the model parameters can be uniquely determined from observed data, and two types of identifiability⁶ are relevant for cure models.

The first type concerns the model structure itself. It asks whether an observed population survival function can be uniquely decomposed into its cure and latency components.

The model is identifiable within function classes P (for incidence) and S (for latency) if

$$(1 - p_1) + p_1 \cdot S_{u,1}(t) = (1 - p_2) + p_2 \cdot S_{u,2}(t) \text{ for all } t \geq 0 \quad (2)$$

implies $p_1 = p_2$ and $S_{u,1} = S_{u,2}$. This type of identifiability depends only on the mathematical form of the model. It determines whether distinct parameter values produce distinct population survival functions, regardless of what data are observed. Hanin and Huang¹³ established the conditions under which various parametric and nonparametric specifications of P and S yield identifiable models.

The authors showed that mixture cure models with common parametric uncured distributions, such as Weibull, log-logistic, gamma, or log-normal, are identifiable in this first sense provided the parametric family of the uncured distribution is itself identifiable. The semiparametric mixture cure model with proportional hazards on the uncured survival (with unspecified baseline hazard) is also identifiable in this sense, typically requiring covariates that distinguish the cure incidence component from the uncured-survival component.

The second type of identifiability is directly relevant to parameter estimation from observed data⁶. It requires that a unique set of parameters maximizes the expected log-likelihood given the

observed data distribution. Patilea and Van Keilegom¹⁴ established conditions for this under parametric incidence models, while Xu and Peng¹⁵ addressed the nonparametric case. Several conditions are relevant for all survival analysis models, but these articles suggest particular follow-up assumptions to achieve desirable properties of cure models. One representation of this type of assumption characterizes the relationship between the event time distribution and the censoring distribution through the so-called sufficient follow-up condition:

$$\tau_{F_0} \leq \tau_G \quad (3)$$

where $\tau_{F_0} = \sup\{t : F_0(t) < 1\}$ denotes the upper support of the event time distribution for susceptibles ($F_0 = 1 - S_u$), and $\tau_G = \sup\{t : G(t) < 1\}$ is the upper support of the censoring distribution G . Condition (3) requires that follow-up extends beyond the time at which all susceptible individuals would have experienced the event.

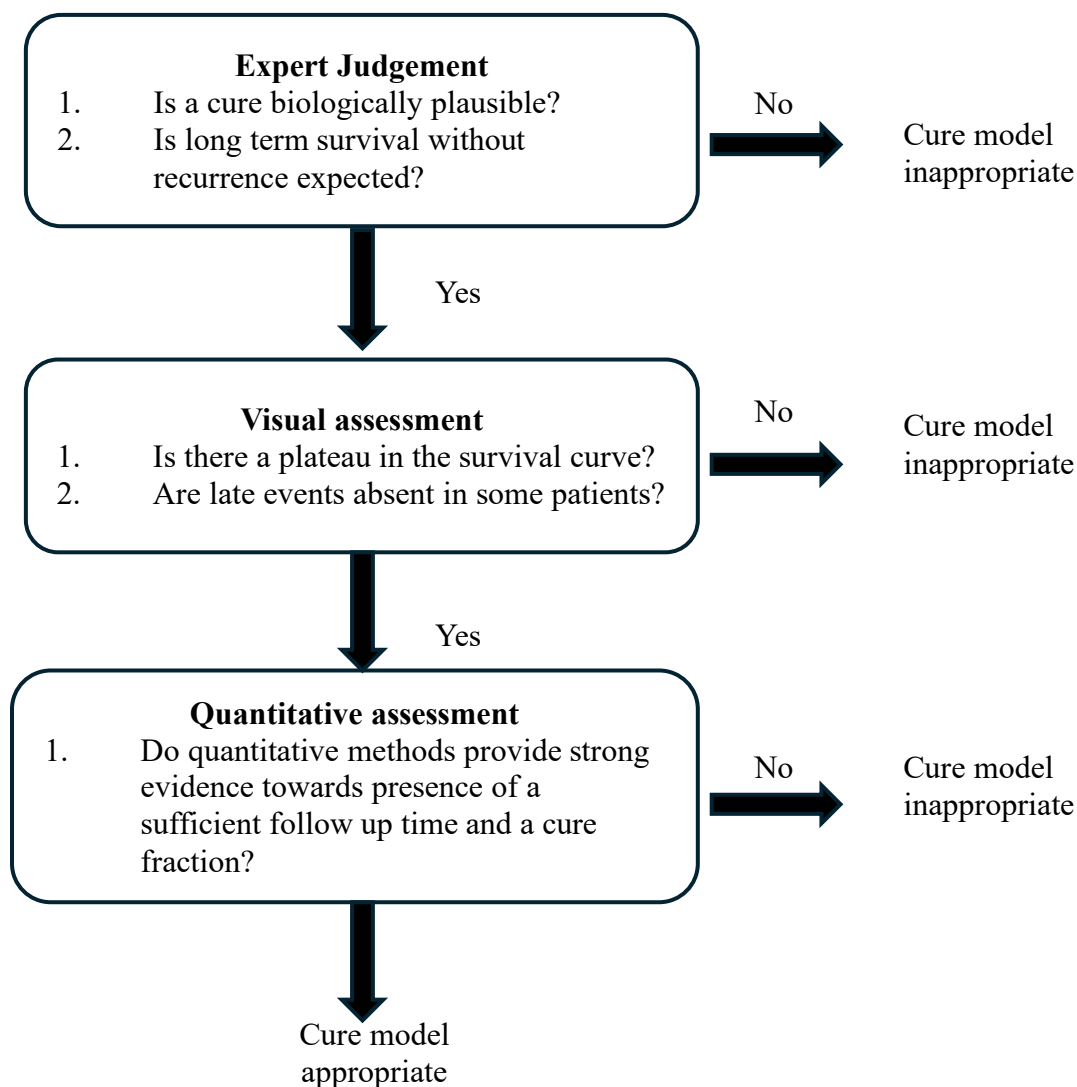
When condition (3) fails, late censored observations could represent either cured individuals or susceptible individuals who would eventually experience the event. These cases are observationally indistinguishable, and the cure fraction cannot be uniquely estimated. Even when the model permits unique decomposition based on the first type of identifiability, estimation requires that condition (3) holds. The methods discussed in this tutorial assess this second form of identifiability, which is essential for valid parameter estimation from observed data.

It is important to distinguish this statistical condition from clinical conventions in which a fixed survival duration (e.g., 5 years post-diagnosis in oncology) is used to designate a patient as "cured." Such conventions reflect domain-specific norms about clinically meaningful long-term survival rather than the identifiability requirement of equation (3). The follow-up needed for valid estimation under a cure model will vary across datasets and must be assessed on a per-study basis using the methods described in Section 2.2.

2.2 | Assessing Cure Model Appropriateness

The following sections describe the assessment process for cure model appropriateness, which proceeds through three stages: clinical judgment, visual assessment, and quantitative evidence. Figure 1 summarizes this process. At each stage, evidence may indicate that a cure model is

inappropriate. In this situations, statisticians frequently apply traditional non-cure models, but authors have recently developed alternative methodology if cure modeling may be reasonable but follow-up seems insufficient. For example, Escobar-Bach and Van Keilegom³⁵ and Escobar-Bach et al.³⁸ developed nonparametric estimators of the cure fraction that remain valid under insufficient follow-up by leveraging extreme-value theory. In this tutorial, we focus on the case where a user is interested in evaluating the appropriateness of a cure model in one group, which has been best studied in the literature, and we discuss multiple groups in the Discussion.



2.2.1 | Step 1: Clinical Judgment

Statisticians should engage clinical experts to evaluate whether cure models could be relevant data models for the specific application. Reviewing prior applications of cure models in comparable datasets can offer guidance, but expert knowledge for the long-term patient trajectories remains critical. In particular, expectations regarding the potential for late failures will inform interpretations for follow-up sufficiency. These discussions can also clarify that true cure may be inappropriate (see Section 2.1.1) and review whether modeling long-term survivorship with cure models may be useful.

2.2.2 | Step 2: Visual Evidence for Cure Fraction and Sufficient Follow-up

The Kaplan-Meier curve provides a nonparametric estimate of $S(t)$ and serves as the primary tool for visual assessment.

Two features in the Kaplan-Meier curve support cure model appropriateness. First, if the curve levels off to a horizontal plateau that does not approach zero, this suggests $\lim_{t \rightarrow \infty} S(t) > 0$, consistent with a non-zero cure fraction. Second, a decreasing event rate prior to the plateau can suggest a population in which susceptible individuals have largely experienced the event, leaving predominantly cured individuals. Recall that each downward step in a Kaplan-Meier curve represents observed events, with height weighted by the number at risk. Because the number at risk decreases over time, late steps are necessarily larger even when events occur at the same or slower pace. The pattern supporting cure model appropriateness is therefore fewer events per unit time near the end of follow-up, not smaller steps.

Visual inspections are inherently subjective, which can lead to low confidence in decisions for cure modeling, especially in situations with limited prior knowledge. To provide further

confirmation when assessing cure model appropriateness, authors have developed quantitative methods as well.

2.2.3 | Step 3: Quantitative Evidence

Since Maller and Zhou introduced the concept of sufficient follow-up in 1992²², there has been significant focus on developing methods to evaluate follow-up adequacy. In their textbook, Maller and Zhou²³ described a two-step hypothesis testing procedure to determine whether a cure model should be applied.

Building on the data setup in Section 2.1, let G denote the cumulative distribution function of the censoring time U_i , assumed independent of the failure times. In the mixture cure model, the population distribution function can be written as $F(t) = p \cdot F_0(t)$, where F_0 is the proper distribution function of the uncured sub-population and $(1 - p)$ remains the cure fraction. Define:

- $Y_{(n)} = \max_{1 \leq i \leq n} Y_i$: the largest observed time
- $Y_{(n)}^* = \max \{Y_i : \delta_i = 1\}$: the largest uncensored observation
- $\tau_{F_0} = \sup \{t : F_0(t) < 1\}$: the right endpoint of the uncured failure-time distribution
- $\tau_G = \sup \{t : G(t) < 1\}$: the right endpoint of the censoring distribution

Testing for a non-zero cure fraction. For the first step, Maller and Zhou²² proposed testing $H_0: p = 1$ versus $H_1: p < 1$, using the estimator $\hat{p}_n = \hat{F}_n(Y_{(n)})$, the Kaplan-Meier estimate evaluated at the largest observed time. They also describe a parametric version using the deviance statistic d_n from a fitted parametric model²⁵. For additional testing methods, see Peng and Yu²⁶.

Testing for sufficient follow-up. For the second step, Maller and Zhou²⁴ established that $\hat{p}_n$ is a consistent estimator of p if and only if $\tau_{F_0} \leq \tau_G$, and proposed testing $H_0: \tau_{F_0} \leq \tau_G$ versus $H_a: \tau_{F_0} > \tau_G$. Under H_0 , follow-up is sufficient. The intuition is that if $\tau_{F_0} > \tau_G$ (insufficient follow-up), then $Y_{(n)} - Y_{(n)}^* \rightarrow 0$ almost surely, so a large value of $Y_{(n)} - Y_{(n)}^*$ provides evidence that $\tau_{F_0} \leq \tau_G$.

Define N_n as the number of uncensored observations in $(2Y_{(n)}^* - Y_{(n)}, Y_{(n)}^*]$. The test statistic is:

$$\hat{\alpha}_n = (1 - N_n/n)^n \quad (5)$$

with H_a rejected (sufficient follow-up accepted) if $\hat{\alpha}_n < 0.05$. Maller and Zhou²³ noted that the $\hat{\alpha}_n$ test can have inflated type I error. To address the problem of type I error inflation, two alternative test statistics were developed by Maller and Zhou²³ and Shen²⁷, known as q_n and $\tilde{\alpha}_n$

respectively. The q_n statistic was initially limited because it required simulation-based percentiles, but this was recently resolved by Maller et al.²⁸. Other authors have also recently developed alternative methodologies within this framework: Xie et al.²⁹ inverted the null and alternative hypotheses for sufficient follow-up, and following the logic from a different framework proposed by Selukar and Othus³⁰ (described next), Yuen & Musta modified the null hypothesis by allowing a user-specified level of uncured to remain at risk at the analysis time.

RECeUS Method. Instead of the two-step framework, Selukar and Othus³⁰ combined testing for a non-zero cure fraction and sufficient follow-up into one procedure. Using the mixture cure framework defined in Section 2.1, let τ denote the maximum observed follow-up time.

Define $\hat{r}_n$, the estimated proportion of uncured among survivors at τ , as:

$$\hat{r}_n = \hat{S}_0(\tau) / [(1 - \hat{p}_n) + \hat{p}_n \cdot \hat{S}_0(\tau)], \quad (7)$$

where the hat notation represents estimates of the corresponding population quantities.

Small values of $\hat{r}_n$ suggest sufficient follow-up for cure modeling. The proposed procedure is:

- (i) Fit parametric cure models and non-cure models. Compare using AIC. If a non-cure model is selected, a cure model is not appropriate.
- (ii) If a cure model is selected, estimate $S(\tau)$, $S_0(\tau)$, and $(1 - p)$.
- (iii) If $(1 - \hat{p}_n)$ is small, then either a cure model is likely not valid or follow-up is likely insufficient.
- (iv) If $(1 - \hat{p}_n)$ is away from 0, estimate r . Small values of $\hat{r}_n$ indicate sufficient follow-up.

Selukar and Othus³⁰ recommend thresholds of $(1 - \hat{p}_n) > 0.025$ and $\hat{r}_n < 0.05$: RECeUS claims a cure model is appropriate when both conditions hold. If one or both conditions fail, RECeUS concludes a cure model is not appropriate.

3 | A WORKED EXAMPLE ON ASSESSING CURE MODEL APPROPRIATENESS

We analyzed data from a phase 3 randomized trial S1203 (ClinicalTrials.gov: NCT01802333) for adult patients with previously untreated acute myeloid leukemia (AML). For illustrative purposes, we summarize results for two study arms: (1) a conventional induction arm with cytarabine and anthracycline followed by consolidation with high-dose cytarabine ("DA"), and (2) induction with idarubicin and high-dose cytarabine followed by consolidation with cytarabine and idarubicin ("IA"). We evaluate cure models motivated by results reported in the primary manuscript by Garcia-Manero et al. (2023). These data have maximum follow-up exceeding 7.3 years since randomization.

We analyze event-free survival (EFS), defined as time to all-cause mortality, relapse after achieving complete remission, or completion of protocol induction/re-induction therapy without achieving complete remission. Both EFS and overall survival (OS) are measured from each patient's time of randomization, and patients without the respective events are censored at their last contact date. We use survival data from the IA arm throughout this worked example in this section to illustrate the cure-model appropriateness assessment.

We provide R code for the worked example in text, and the code is also available at <https://github.com/GeethanjaleeM/cureAssess>.

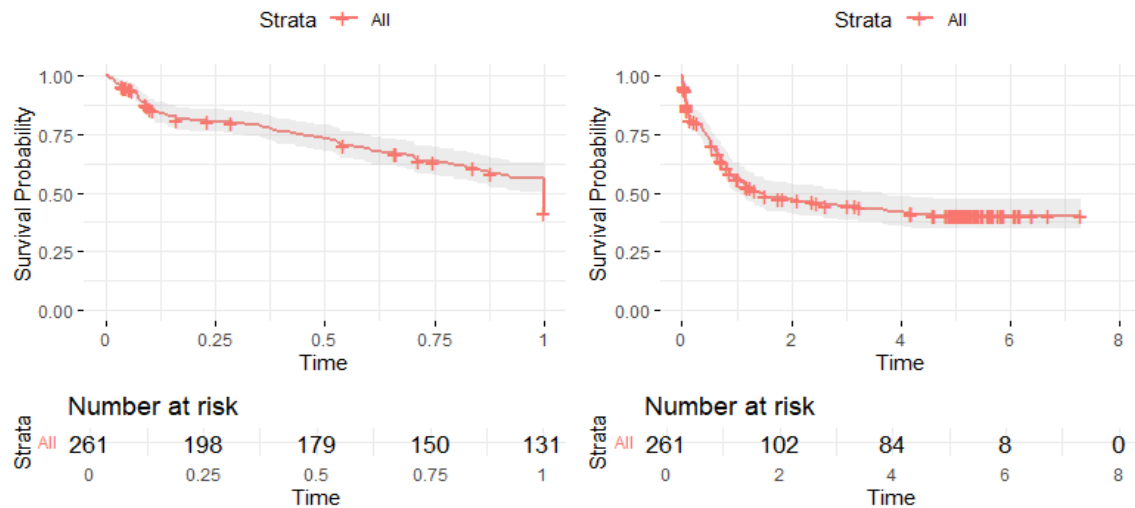
The following steps illustrate the assessment process.

Step 1: Clinical Rationale for a Cured Subpopulation

In the context of AML, approximately 22.6% of adult AML patients survive to five years. While late relapses are possible, 5-year survivors are typically considered cured³¹. This provides evidence for a cured fraction in the population considered in this study.

Step 2: Visual Evidence

[FIGURE 2: (left) KM estimates of EFS for the IA arm after artificially restricting follow-up time to 1 year; (right) KM estimates of EFS for the IA arm for the total follow-up time]



The right panel of Figure 2 shows the Kaplan-Meier curve for the IA arm using all available follow-up (more than six years). We observe no events for over a year before the end of follow-up, producing a long horizontal plateau, and the rate of events appears lower after year four compared to earlier. To illustrate what inadequate follow-up could look like by contrast, the left panel shows the same data after artificially restricting follow-up to one year. In this restricted view, events occur shortly before one year, the curve continues to step downward up to the end of the period, and no horizontal plateau is visible.

From this visual comparison, the right panel suggests that patients remaining at risk after six years are unlikely to experience an EFS event with further follow-up, supporting the plausibility of a cure model. Conversely, an observer would likely be skeptical of a similar claim from the left panel alone. We therefore conclude that a cure model may be appropriate when considering the full available follow-up but would likely not be appropriate if the study follow-up was restricted to one year.

Step 3: Quantitative Evidence

For illustrative purposes, we evaluate cure model appropriateness with the $\hat{\alpha}_n$ test²⁴ and the RECeUS method³⁰. Maller & Zhou's test of sufficient follow-up indicates strong evidence ($p < 0.05$) for the IA arm. The RECeUS method with a Weibull model also suggests strong evidence for the IA arm for EFS.

Model Selection. We fit a class of parametric cure and non-cure models and use AIC to select an appropriate model. (Importantly, this model for assessing cure model appropriateness can differ from the analysis model.)

Table 1: Comparing parametric mixture cure models and non-cure models

Model	IA Arm AIC
Exponential non-cure	685.27
Exponential cure	545.44
Weibull non-cure	574.58
Weibull cure	537.99*
Gamma non-cure	587.22
Gamma cure	539.54
Log-logistic non-cure	558.05
Log-logistic cure	538.27

*Lowest AIC

The Weibull cure model has the smallest AIC value for the IA arm, so we proceed to estimate $(1 - \hat{p}_n)$ and $\hat{r}_n$.

R Code for Fitting Weibull Cure Model:

```
# Load required package
library(flexsurvcure)

# Fit Weibull cure model for IA arm
fit_weibull_cure <- flexsurvcure(
  Surv(efs_time, efs_event) ~ 1,
  data = ia_arm,
  dist = "weibull"
)

# View results
summary(fit_weibull_cure)
```

Output:

```
Estimates:
      est      L95%      U95%      se
```

```
theta  0.3976  0.3346  0.4642    NA
shape  0.8133  0.7052  0.9379  0.0592
scale  0.8052  0.6362  1.0191  0.0968
```

```
N = 261, Events: 142, Censored: 119
Total time at risk: 579.2279
Log-likelihood = -265.9932, df = 3
AIC = 537.9865
```

RECeUS Calculations:

From the fitted model:

- $(1 - \hat{p}_n) = 0.3976$ (estimated cure fraction)
- $\tau = 2659 \text{ days} / 365.25 = 7.28 \text{ years}$ (maximum follow-up)
- $\text{shape} = 0.8133, \text{scale} = 0.8052$

Calculate the survival function for susceptibles at τ :

$$\hat{S}_0(\tau) = \exp[-(\tau/\text{scale})^{\text{shape}}] = \exp[-(7.28/0.8052)^{0.8133}] = 0.00249$$

Calculate the population survival at τ :

$$\hat{S}(\tau) = (1 - \hat{p}_n) + \hat{p}_n \times \hat{S}_0(\tau) = 0.3976 + (0.6024)(0.00249) = 0.3991$$

Calculate the ratio:

$$\hat{r}_n = \hat{S}_0(\tau) / \hat{S}(\tau) = 0.00249 / 0.3991 = 0.00625$$

R Code for RECeUS Calculation:

```
# Extract parameters
theta <- coef(fit_weibull_cure)['theta'] # cure fraction
shape <- coef(fit_weibull_cure)['shape']
scale <- coef(fit_weibull_cure)['scale']

# Maximum follow-up time in years
tau <- max(ia_arm$efs_time) / 365.25

# Calculate S0(tau) for Weibull distribution
S0_tau <- exp(-(tau/scale)^shape)

# Calculate r_hat
r_hat <- S0_tau / (theta + (1 - theta) * S0_tau)
```

```
# Check thresholds
cat('Cure fraction =', round(theta, 4), '\n')
cat('r_hat =', round(r_hat, 4), '\n')
cat('Cure model appropriate:', theta > 0.025 & r_hat < 0.05, '\n')
```

Decision:

Since $(1 - \hat{p}_n) = 0.3976 > 0.025$ and $\hat{r}_n = 0.00625 < 0.05$, both conditions are satisfied. Therefore, RECeUS confirms that a cure model is appropriate for this dataset.

4 | OTHER PRACTICAL CONSIDERATIONS FOR APPROPRIATE APPLICATION OF CURE MODELS: BMTCT DATA EXAMPLES

To share broader considerations for appropriately applying cure models, we applied the assessment framework to a series of datasets from Bone Marrow Transplantation & Cellular Therapy (BMTCT) research at St. Jude Children's Research Hospital. These included both prospective clinical trials and retrospective analyses. The goals of this work were to evaluate potential cure model appropriateness across (1) different patient populations and (2) varying follow-up procedures. Goal (1) addresses different patient populations: we included BMTCT cohorts with different expected prognoses (e.g., patients in complete remission [CR] at the time of transplantation are expected to have better prognosis than patients with active disease at the time of transplantation). Goal (2) addresses varying follow-up procedures by including both prospective trials (with active, protocol-specified monitoring) and retrospective analyses (where follow-up depends on routine clinical documentation rather than protocol-specified visits), allowing the appropriateness framework to be evaluated across the kinds of follow-up regimes investigators encounter in practice.

The prospective datasets comprised two studies with multiple prespecified cohorts: HAPNK1 (phase 2 study in high-risk hematological malignancies; CR cohort, $n = 53$; active disease cohort, $n = 19$) and HAP2HCT (phase 1/2 study; phase 2 patients at dose levels 3-4, $n = 48$).

The retrospective datasets comprised two analyses, also with multiple prespecified cohorts: HCTRETRO (patients receiving ≥ 2 HCTs; 2nd HCT, $n = 106$; > 2 nd HCT, $n = 13$) and Refractory at HCT (patients with refractory disease at transplant, $n = 129$).

The complete table of cure model appropriateness assessments for all datasets is provided in the **Supplementary Materials** (Table S1). Here we present one illustrative example.

Example: HCTRETRO 2nd HCT Cohort

This cohort ($n = 106$) had median follow-up of 0.84 years and maximum follow-up of 17.7 years. The Kaplan-Meier estimate at maximum follow-up was 0.257. Visual inspection showed a plateau in the KM curve consistent with long-term survivors, and AIC comparison selected the Weibull cure model. Then, RECeUS assessment confirmed cure model appropriateness. This example illustrates how retrospective data with extended follow-up can support cure model application despite the lack of protocol-specified follow-up.

5 | DISCUSSION

It is important for cure models to be clinically meaningful and appropriately tailored to the specific data being analyzed. While potentially useful tools, applying cure models when conditions are not met can lead to misleading results. In this tutorial, we provide guidance to assess the appropriateness of cure models through a combination of expert judgment, visual assessment, and quantitative evaluation. First, expert knowledge determines whether a cure is biologically or clinically plausible. Next, visual inspection of the Kaplan-Meier curve can reveal a plateau that does not approach a survival probability of zero, suggesting the presence of long-term survivors. Finally, statistical methods can provide quantitative evidence to confirm the appropriateness of cure models.

Recent work by Selukar and Othus³⁰ and Yuen and Musta³⁶ proposed that cure models may still provide useful inference even when the classical sufficient follow-up condition $\tau_{F_0} \leq \tau_G$ is not strictly satisfied. Specifically, these methods allow estimation when a small but non-negligible proportion of susceptible individuals remain at risk at the end of follow-up. While simulation studies support this approach, the theoretical justification remains an area of ongoing research. There is no universally accepted threshold for how much the condition can be relaxed while still obtaining valid inference.

This tutorial evaluates one group at a time, as this has been the case for each method proposed thus far. Kouadio³⁷ et al. investigated cure model appropriateness for randomized controlled trials and suggest that appropriateness in at least one of two arms could be sufficient for applying cure models. However, in their general proofs for large-sample properties of cure models, Patilea and Van Keilegom¹⁴ assume that cure model appropriateness should hold for each element of the covariate space, so considerations for models with more than two groups may become more difficult. Evaluating appropriateness for each possible group may be challenging due to multiplicity and sample size limitations, so special considerations may be required.

Through our analyses, we identified other practical considerations for applying cure models. Cure model appropriateness cannot be predicted solely by the quantity of follow-up: We identified scenarios where high-risk populations with suitable follow-up have evidence to potentially be appropriately modeled with cure models, while lower-risk populations may not, even with longer quantity of follow-up. We also observed data quality issues for cure modeling when follow-up transitioned from active trial monitoring to passive follow-up.

6 | CONCLUSION

In summary, while cure models offer valuable insights for clinical trials, their implementation requires careful attention to data and study considerations such as follow-up time, data quality, and model appropriateness. By ensuring clinical relevance and assessing these factors using the systematic procedure described in this tutorial, future research can enhance the utility and reliability of cure models, paving the way for more effective research and improved patient outcomes.

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SUPPLEMENTARY MATERIALS

Table S1: Evaluating Cure Model Appropriateness for BMTCT Datasets

The complete table containing all BMTCT dataset assessments is provided below. Columns include: Dataset, Cohort, Sample size, Median time, Maximum follow-up time (years), Kaplan-Meier estimate at maximum follow-up, Visual evidence for cure (Yes/No), Visual evidence for sufficient follow-up (Yes/No), Cure model selected by AIC (Yes/No), Best-fitting model name, and RECeUS assessment.

Table S1. Evaluating cure model appropriateness for BMTCT datasets

Dataset	Cohort	Sample size	*Median time	Median time from HCT to event or censoring (years)	Maximum follow-up time (years)	Kaplan-Meier estimate at maximum follow-up time (years)	Visual evidence for cure (Yes/No)	Visual evidence for sufficient follow-up (Yes/No)	Cure model selected (Yes/No)	AIC best-fitting model (name of the model)	Cure model appropriate by RECeUS
HAPNK1	CR	53	3.5702	NA	8.6078	0.5971	Yes	Yes	No	Lognormal	No
HAPNK1	Active disease	19	1.2841	0.4791	7.8713	0.1973	Yes	Yes	Yes	Exponential cure	Yes
HAP2HCT_ALL	Phase 2	48	1.0021	1.5852	1.6181	0.2771	No	No	No	Exponential	No
HAP2HCT_1yr	Phase 2	48	0.9993	NA	1	0.79166	Yes	Yes	Yes	Weibull cure	Yes
HCTRETRO	2 nd HCT	106	0.8419	0.5010	17.7248	0.2570	Yes	Yes	Yes	Weibull cure	Yes
HCTRETRO_10yrs	2 nd HCT	106	0.8419	0.5010	10	0.2570	Yes	Yes	Yes	Weibull cure	Yes
HCTRETRO	>2 nd HCT	13	1.4428	0.7420	11.0445	0.1538	~Yes (Not enou)	~Yes (Not enou)	Yes	Lognormal cure	No (Marginal)

							gh data)	gh data)			Decisi on)
HCTRETRO _10yrs	>2 nd HCT	13	1.44 28	0.742 0	10	0.153 8	~Yes (Not enou gh data)	~Yes (Not enou gh data)	No	Lognor mal	No
Refractory at HCT	All	129	0.30 66	0.306 6	29.66 46	0.145 92	Yes	Yes	Yes	Loglogi stic cure	No (Margin al Decisi on)
Refractory at HCT_5yrs	All	129	0.30 66	0.306 6	5	0.162 8	Yes	Yes	Yes	Weibull cure	Yes
SCDHCT	MSD	9	2.62 56	NA	3.047 2	0.888 9	~Yes (Not enou gh data)	~Yes (Not enou gh data)	Yes	Expone ntial cure	Yes
SCDHCT_3 yrs	MSD	9	2.62 56	NA	3	0.888 9	~Yes (Not enou gh data)	~Yes (Not enou gh data)	Yes	Expone ntial cure	Yes
SCDHCT	Hapl o	11	2.16 29	NA	3.118 4	0.818 2	~Yes (Not enou gh data)	~Yes (Not enou gh data)	Yes	Weibull cure	Yes

R Code for Complete RECeUS Assessment Function:

```
# RECeUS Assessment Function
receus_assessment <- function(surv_data, time_var, event_var, dist =
'weibull') {
  library(flexsurvcure)
  library(survival)

  # Fit cure and non-cure models
  formula <- as.formula(paste0('Surv(', time_var, ', ', event_var, ') ~ 1'))

  fit_cure <- flexsurvcure(formula, data = surv_data, dist = dist)
  fit_noncure <- flexsurvreg(formula, data = surv_data, dist = dist)

  # Compare AIC
```

```

aic_cure <- AIC(fit_cure)
aic_noncure <- AIC(fit_noncure)
cure_selected <- aic_cure < aic_noncure

if (!cure_selected) {
  return(list(
    cure_selected = FALSE,
    appropriate = FALSE,
    message = 'Non-cure model selected by AIC'
  ))
}

# Extract parameters (theta = cure fraction in flexsurvcure)
theta <- coef(fit_cure)['theta']
shape <- coef(fit_cure)['shape']
scale <- coef(fit_cure)['scale']

# Maximum follow-up
tau <- max(surv_data[[time_var]])

# Calculate S0(tau) for Weibull
S0_tau <- exp(-(tau/scale)^shape)

# Calculate r_hat
r_hat <- S0_tau / (theta + (1 - theta) * S0_tau)

# Assessment
appropriate <- (theta > 0.025) & (r_hat < 0.05)

return(list(
  cure_selected = TRUE,
  aic_cure = aic_cure,
  aic_noncure = aic_noncure,
  cure_fraction = theta,
  r_hat = r_hat,
  appropriate = appropriate
))
}

# Example usage:
# result <- receus_assessment(my_data, 'time', 'event', dist = 'weibull')
# print(result)

```